
Constitutional AI: Harmlessness from AI Feedback

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Abstract

As AI systems become more capable, we would like to enlist their help to supervise other AIs. We experiment with methods for training a harmless AI assistant through self-improvement, without any human labels identifying harmful outputs. The only human oversight is provided through a list of rules or principles, and so we refer to the method as ‘Constitutional AI’. The process involves both a supervised learning and a reinforcement learning phase. In the supervised phase we sample from an initial model, then generate self-critiques and revisions, and then finetune the original model on revised responses. In the RL phase, we sample from the finetuned model, use a model to evaluate which of the two samples is better, and then train a preference model from this dataset of AI preferences. We then train with RL using the preference model as the reward signal, i.e. we use ‘RL from AI Feedback’ (RLAIF). As a result we are able to train a harmless but non-evasive AI assistant that engages with harmful queries by explaining its objections to them. Both the SL and RL methods can leverage chain-of-thought style reasoning to improve the human-judged performance and transparency of AI decision making. These methods make it possible to control AI behavior more precisely and with far fewer human labels.

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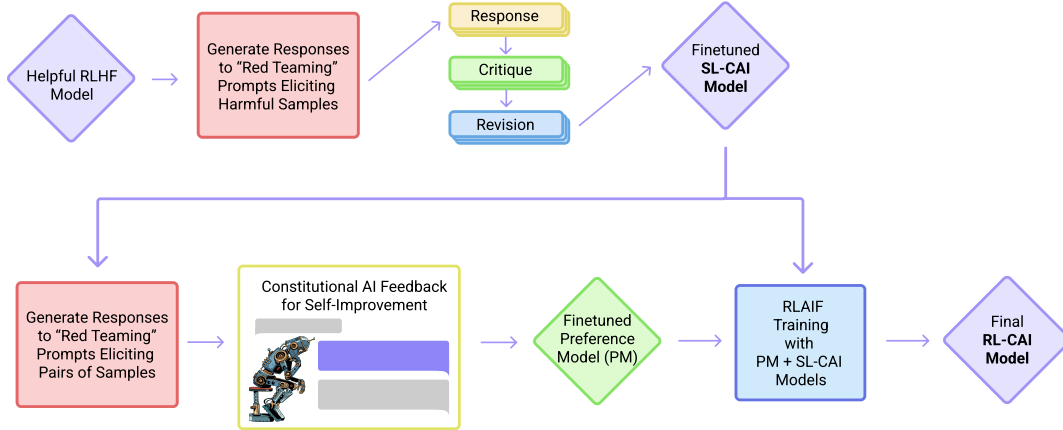


Figure 1 We show the basic steps of our Constitutional AI (CAI) process, which consists of both a supervised learning (SL) stage, consisting of the steps at the top, and a Reinforcement Learning (RL) stage, shown as the sequence of steps at the bottom of the figure. Both the critiques and the AI feedback are steered by a small set of principles drawn from a ‘constitution’. The supervised stage significantly improves the initial model, and gives some control over the initial behavior at the start of the RL phase, addressing potential exploration problems. The RL stage significantly improves performance and reliability.

1 Introduction

We would like to train AI systems that remain helpful, honest, and harmless, even as some AI capabilities reach or exceed human-level performance. This suggests that we will need to develop techniques that do not rely on humans to supervise all aspects of AI behavior, and that can be used to automatically test and enhance robustness to harmful behaviors. We also aim to develop methods that encode desirable AI behavior in a simple and transparent form, and that make it easier to understand and evaluate AI decision making.

In this paper we develop a method we refer to as Constitutional AI (CAI), depicted in Figure 1, and use it to train a non-evasive and relatively harmless AI assistant, *without any human feedback labels for harms*. The method therefore improves upon, and partially replaces reinforcement learning from human feedback [Christiano et al., 2017]. The new assistant ‘RL-CAI’ is preferred by crowdworkers over those trained with previously collected [Bai et al., 2022, Ganguli et al., 2022] human feedback labels for harmfulness. We chose the term ‘constitutional’ because we are able to train less harmful systems entirely through the specification of a short list of principles or instructions, i.e. a constitution. But we are also employing this terminology to emphasize that when developing and deploying a general AI system, we cannot avoid choosing some set of principles to govern it, even if they remain hidden or implicit.

Our motivations for developing this technique were: (1) to study simple possibilities for using AI systems to help supervise other AIs, and thus *scale supervision*, (2) to improve on our prior work training a harmless AI assistant by *eliminating evasive responses*, reducing tension¹ [Bai et al., 2022, Glaese et al., 2022] between helpfulness and harmlessness and encouraging the AI to explain its objections to harmful requests, (3) to make the principles governing AI behavior, and their implementation, more transparent, and (4) to reduce iteration time by obviating the need to collect new human feedback labels when altering the objective. Let us discuss these motivations in more detail.

1.1 Motivations

Scaling Supervision

We use the term ‘Scaling Supervision’ for techniques that leverage AI to help humans to more efficiently supervise AI, making it possible to train systems to behave in desirable ways (e.g. to be helpful, honest, and

¹That is, helpfulness tends to increase harmfulness, since models are willing to obey pernicious requests, and conversely models trained to be harmless tend to be more evasive and generally less helpful. By harmfulness we include both a variety of forms of harm to the user and responses that help the user to achieve harmful aims. See [Bai et al., 2022, Ganguli et al., 2022] for more discussion of our operational definitions of helpful and harmless.

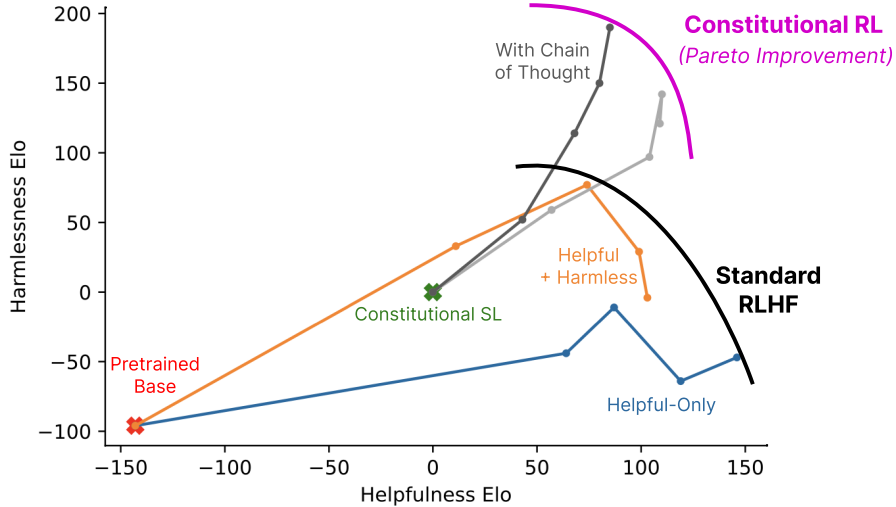


Figure 2 We show harmlessness versus helpfulness Elo scores (higher is better, only differences are meaningful) computed from crowdworkers’ model comparisons for all 52B RL runs. Points further to the right are later steps in RL training. The Helpful and HH models were trained with human feedback as in [Bai et al., 2022], and exhibit a tradeoff between helpfulness and harmlessness. The RL-CAI models trained with AI feedback learn to be less harmful at a given level of helpfulness. The crowdworkers evaluating these models were instructed to prefer less evasive responses when both responses were equally harmless; this is why the human feedback-trained Helpful and HH models do not differ more in their harmlessness scores. Error bars are visible in Figure 3 but are suppressed here for clarity.

harmless [Askell et al., 2021]) with a smaller quantity of higher quality human supervision. There are several reasons why this may be useful:

- AI supervision may be more efficient than collecting human feedback. It allows us to focus more on providing a small amount of legible, focused, high-quality oversight. There may also be ways for humans and AI systems to collaborate [Bowman et al., 2022] to provide better supervision than either can provide alone.
- AI systems can already perform some tasks at or beyond human level (e.g. [Silver et al., 2017]), and over time more examples are likely to emerge. We need to develop methods now that can provide oversight for these powerful AI systems, and scaling supervision may be one possibility, if the capability level of the supervisor can scale proportionally with the capabilities of the actor, *and* the supervisor remains aligned with our intended goals and constraints.

That said, scaling supervision could also have downsides and dangers, since it means further automating (and quite possibly obscuring) decision making. As we discuss below, our constitutional approach leverages chain-of-thought reasoning [Nye et al., 2021, Wei et al., 2022] to make decision making more legible.

In a certain sense, work on reinforcement learning from human feedback [Stiennon et al., 2020, Bai et al., 2022, Ouyang et al., 2022] has already taken a step in the direction of scaled supervision, since the reward signal in RL actually comes from an AI preference model (PM) rather than from immediate human oversight. However, RLHF typically uses tens of thousands of human preference labels.

Here, we will test methods that reduce human input to an extreme, in order to study their viability. We will finetune AI models to be harmless using only of order 10^2 simple principles, stated in natural language.

²These principles were chosen in a fairly ad hoc and iterative way for research purposes. In the future, we believe such principles should be redeveloped and refined by a larger set of stakeholders, and that they should also be adapted depending on the intended usage and location in which the model may be deployed. Since such a small number of bits of information are involved in these principles, it’s worth studying these bits carefully.